

Genome-based reclassification of *Terribacillus goriensis* (Kim *et al.* 2007) Krishnamurthi and Chakrabarti 2009 as a later heterotypic synonym of *Terribacillus saccharophilus* An *et al.* 2007

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Abstract

The present study used whole-genome data to clarify the taxonomic assignment of two closely related *Terribacillus* species. The genus *Terribacillus* includes four species with validly published names, *Terribacillus aidingensis*, *T. goriensis*, *T. halophilus* and *T. saccharophilus*. The type species *T. saccharophilus* 002-048^T (=IAM 15309^T=KCTC 13936^T) was isolated from the soil in Japan, while *T. goriensis* CL-GR16^T (=KCCM 42329^T=DSM 18252^T) was isolated from the coastal water off the east coast of Korea. Both strains have similar phenotypic and chemotaxonomic characteristics. Anteiso-C_{15:0} was the major fatty acid, followed by anteiso-C_{17:0} and iso-C_{15:0}. There were similar major menaquinones (MK-7). As determined by whole-genome sequencing, the DNA G+C content of 42.53 and 42.66 mol% was found for *T. saccharophilus* and *T. goriensis*, respectively, and they were 99.86% identical in their 16S rRNA gene sequences (1,421 bp). Genomic comparisons showed the average nucleotide identity was 98.29% and 84.80% for digital DNA–DNA hybridization. These values exceeded the threshold values for bacterial species delineation. Using the Type (Strain) Genome Server, both species were identified as *T. saccharophilus*. Phylogenetic analysis based on both 16S rRNA gene sequences and whole-genome sequences showed that they form a separate cluster apart from members of other genera. Based on the combined evidence, *Terribacillus goriensis* Kim *et al.* (*Int J Syst Evol Microbiol* 2007;57:1554–1560), Krishnamurthi and Chakrabarti (*Int J Syst Evol Microbiol* 2008;58:2287–2291) is proposed to be a later heterotypic synonym of An *et al.* (*Int J Syst Evol Microbiol* 2007;57:51–55).

The genus *Terribacillus* was first described by An *et al.* [1]. Currently, there are four species with validly published names in this genus on the List of Prokaryotic Names with Standing in Nomenclature (<https://lpsn.dsmz.de/genus/terribacillus>, October 2024). *T. saccharophilus* and *T. halophilus* were isolated from the field soil in Japan [1]. *Pelagibacillus goriensis* was isolated from the coastal water off the east coast of Korea [2]. The bacterium was later renamed *T. goriensis* [3]. Finally, *T. aidingensis* was isolated from the sediments of Aiding salt lake in the Xinjiang region of China [4]. The type strain *T. saccharophilus* 002-051^T (=IAM 15310^T=KCTC 13937^T) is a Gram-stain-positive, aerobic, non-motile, spore-forming rod, capable of growing at 5–45 °C and 0–16% (w/v) NaCl (optimum, 1–5% NaCl) [1]. The type strain *T. goriensis* CL-GR16^T (=KCCM 42329^T=DSM 18252^T) is a Gram-stain-positive, aerobic, motile, spore-forming rod, capable of growing at 15–43 °C and 0–14% (w/v) NaCl (optimum, 0–2% NaCl) [2]. Both strains contained MK-7 as the major menaquinone and anteiso-C_{15:0} as the major fatty acid. Both bacteria were compared in the article by Krishnamurthi and Chakrabarti [3]. However, their relationship was not properly delineated.

The whole-genome sequences of strains within the genus *Terribacillus* and closely related genera were obtained from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) database (<https://www.bv-brc.org/>). The genomes were annotated using the Rapid Annotation using Subsystem Technology tool kit [5], which can be found at BV-BRC. Codon Tree [6] was used to construct a phylogenetic tree (Fig. 1). Genomes were compared to each other using average nucleotide identity (ANI) with default parameters (<https://www.ezbiocloud.net/tools/ani/>). In addition, digital DNA–DNA hybridization (dDDH) values were also calculated using the Type (Strain) Genome Server (TYGS) with default parameters [7]. 16S rRNA gene sequences from the members of the genus *Terribacillus* and closely related genera were retrieved using the EzBioCloud-server (<https://www.ezbiocloud.net/>) [8], and sequences were aligned using Clustal W [9]. A phylogenetic analysis was then done using MEGA X [10],

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Abbreviations: ANI, average nucleotide identity; dDDH, digital DNA–DNA hybridization; ML, maximum-likelihood.

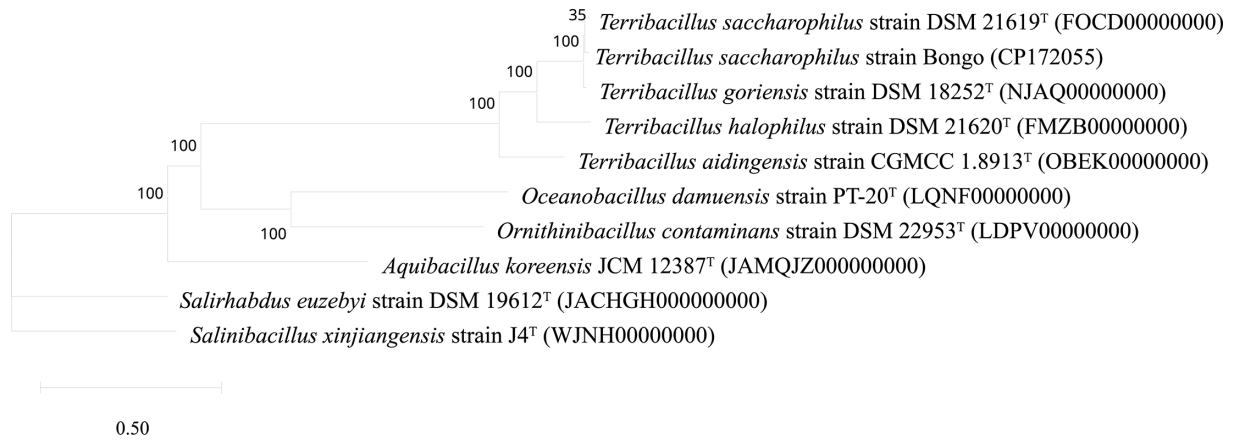


Fig. 1. A phylogenetic tree based on the genomes of *T. goriensis* CL-GR16^T and *T. saccharophilus* 002-051^T and their nearest phylogenetic relatives. The tree was constructed using Codon Tree. Database accession numbers are given in parentheses.

and a phylogenetic tree was constructed using the maximum-likelihood (ML) algorithm [11] with the Kimura two-parameter model used to calculate genetic distances [12]. Bootstrap values were calculated based on 1,000 replications [13]. The phylogenetic analyses using the ML method are shown in Fig. 2.

The genome information showed a DNA G+C content of 42.53 and 42.66 mol% for *T. saccharophilus* and *T. goriensis*, respectively (Table 1). When the 16S rRNA gene sequences (1,421 bp) were compared, a value of 99.86% identity was obtained,

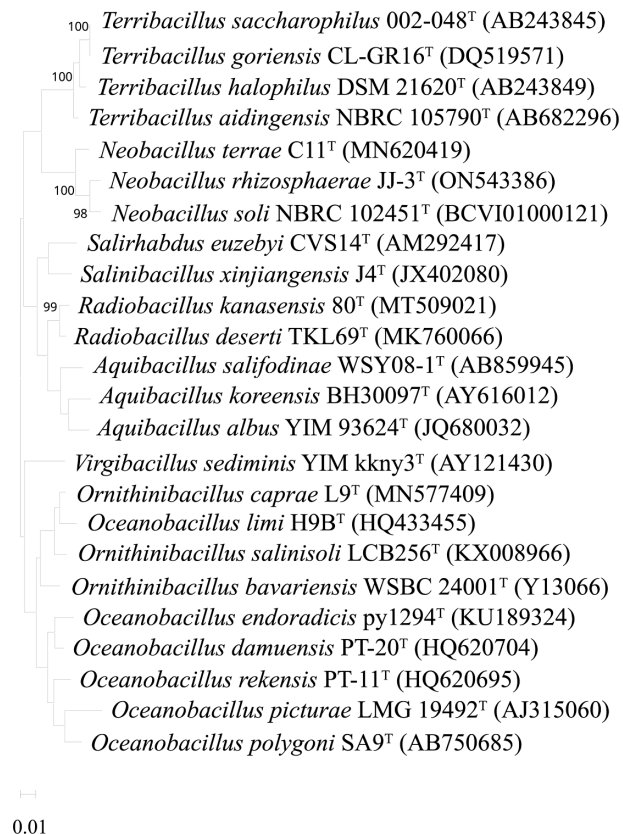


Fig. 2. Phylogenetic tree based on 16S rDNA sequences showing the nearest phylogenetic relatives of *T. goriensis* CL-GR16^T and *T. saccharophilus* 002-051^T. The tree was constructed using the ML method. Bootstrap values (over 90%), expressed as percentages of 1,000 replications, are given at branching points. Database accession numbers are given in parentheses.

Table 1. Differential characteristics. 1. *T. goriensis* CL-GR16^T 2. *T. saccharophilus* 002-051^T. Data were taken from [3] +, Positive; –, negative

Characteristic	1	2
Endospores	+	+
Motility	+	–
Maximum NaCl for growth (% w/v)	14	16
Oxidase	–	–
Urease	–	–
Indole	–	–
H ₂ S Production	–	–
Major menaquinone	MK-7	MK-7
Utilization of:		
L- Arabinose	–	–
Citrate	+	–
Xylose	+	–
D-Fructose	–	+
D-Glucose	–	+
Glycerol	–	+
Mannitol	–	+
Sucrose	–	+
Galactose	+	+
Raffinose	+	+
Trehalose	+	+
Genomic features: DNA G+C content (mol %)	42.66	42.53
Size (Mb)	3.6	3.6
CDS	3737	3790
16S rRNA similarity (%)	99.86	99.86

indicating a very close relationship. The phylogenetic tree based on the 16S rRNA genes showed a very close relationship between *T. saccharophilus* and *T. goriensis* (gene sequence similarity value of 99.86%) forming a distinct monophyletic clade with a gene sequence similarity value of 99.86 % clade separated from other genera (Fig. 2). This value exceeds 98.7 % threshold value used in the assignment of strains to a species. This relationship confirmed the findings in the study by Krishnamurthi and Chakrabarti [3].

To further resolve the taxonomic assignment, ANI was utilized. It has been shown that ANI represents a good measure of both the evolutionary and genetic distance between bacterial strains [14]. For ANI, the proposed species boundary cut-off is 95–96% [15]. When the two genomes were compared by ANI (<https://www.ezbiocloud.net/tools/ani>), a value of 98.29% was obtained. This value was well above the cutoff value normally used to delineate separate species of bacteria. Furthermore, this high level of relatedness was further confirmed using TYGS that showed that both genomes were derived from the same species with a DDH value of 84.80%. In addition to the 16S rRNA analysis, the genome-related measures used for species delineation, such as ANI and dDDH, exceeded the accepted thresholds for species, which are <98.7 for 16S rRNA [16], 95% for ANI [15] and <70% for dDDH [17].

The high similarity values of genome-related indices were consistent with the similar genome sizes and G+C content percentages (Table 1). Core genome phylogeny was performed. A phylogenetic tree was then reconstructed using Codon Tree (Fig. 1), which showed a very close relationship between the two species. Together, the ANI, TYGS and the phylogenetic and phylogenomic trees clearly demonstrate that *T. saccharophilus* 002-051^T and *T. goriensis* CL-GR16^T belong to the same species.

Considering the high level of genetic similarity, it is not surprising that biochemical and physiological traits of the strains show many similarities with differences attributed to intraspecies differences due to environmental sources.

As per the priority of prokaryotic names governed by the International Code of Nomenclature of Prokaryotes [18], when species are united, the earliest validly published name is used for the union. Therefore, *T. goriensis* [2, 3] is a later heterotypic synonym of [1] An et al. 2007.

EMENDED DESCRIPTION OF *T. SACCHAROPHILUS*

The description is as before [1, 2] with the following modifications.

The type strain is DSM 21619 (=IAM 15309=JCM 21759=KCTC 13936=002-048), and strain DSM 18252 (=KCCM 42329=CL-GR16) is an additional strain. The additional strain of *T. saccharophilus* grows between 15 and 43 °C, pH 5.5–9.0. Optimum growth in NaCl is 0–2% (w/v) with a range of 0–14%. The motility between the two strains is variable, as is citrate, fructose, glucose, glycerol, mannitol, sucrose and xylose utilization, as well as acetoin production. Galactose, raffinose and trehalose are utilized. However, arabinose is not. Both strains are positive for catalase activity and gelatin hydrolysis and negative for oxidase activity, anaerobic growth, reduction in nitrate, H₂S production, arginine dihydrolase, lysine decarboxylase, tryptophan deaminase, urease and indole production.

The emended DNA G+C content (mol %) is 42.53–42.66. The genome sequence is deposited in NCBI as GCF_900110015.1. The genome size is 3.6 Mb.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

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